

THE GRAIL COLLECTOR

Roadmap Rev I — Completion Status (updated, Rev 4.2)

Item-by-item status of the Design and Simulation Roadmap (Rev I, corrected) after the CFD phase and Stages 3, 5, 6 and 7 (ANN, NSGA-II, system model, annual analysis). The roadmap itself is not rewritten; this document records what was done, how, and what remains.

Item	Value
Roadmap version	Rev I (corrected), 64 pages
Phases closed	CFD (Sections 4–8, 12, 15); ANN (10); system model and annual analysis (11, via Python, GNU Octave, Scilab Xcos and MATLAB/Simulink, all agreeing)
Not done	Section 9 topology optimisation (replaced by NSGA-II parameter optimisation on the ANN surrogate)
Final dataset	GRAIL_CFD_dataset_FINAL_rev4.2.csv — 300 cases, audited (Rev 4.1 data, three columns renamed)
Main deviation	COMSOL 6.2 → OpenFOAM v1912 (3-D) + GRail-CHT reduced-order conjugate model
Experimental validation	None of the present collector. V5 COMPLETE (method/component level): 16 papers screened; method checks against Del Col 2013 (roll-bond: a1, a2 within uncertainty, η_0 +6–8 %) and Beikircher 2015 (TIM front loss within measured range; VIP rear loss optimistic)

Contents

1. Section-by-section status

Roadmap section	Status	What was done / why not
1–3 Thesis, gaps, prior art	REFERENCE	Background; not a work item
4 Design parameter set	USED	Groups A, B, E sampled: G_T , T_{in} , T_{amb} , v_{wind} , $\dot{m}$, bridge, G , λ_G , $f_{interdig}$
5 Architecture, BOM, properties	USED	Material database recorded; V6 supplier datasheets not obtained
6 Fusion 360 modelling	DONE	Grail_Collector_2.step, 129 solids; Gate A 12/12 invariants PASS
7 Fusion → COMSOL	REPLACED	Direct NURBS evaluation and structured hex meshing for OpenFOAM
8.1–8.6 CFD methodology	DONE (deviation)	OpenFOAM simpleFoam + chtMultiRegionSimpleFoam (frozenFlow); GRAIL-CHT for the campaign
8.7 Validation	PARTIAL	Code-to-code (3-D conjugate) and code-to-correlation (H1 Nu, HWB); Gunjo 2017 rejected; Sun et al. not obtained
8.8 Campaign	DONE	300 LHS cases, 0 rejected (Rev 4.1); log fields as specified
9 Topology optimisation	NOT DONE	Replaced by NSGA-II parameter optimisation on the ANN surrogate (Stage 5); 8 optima re-run in GRAIL-CHT
10 ANN / hybrid ANN	DONE	Stage 3: 10-seed MLP ensemble, test R^2 0.9994–0.9998 (η), GPR benchmark, SHAP
11 MATLAB/Simulink	DONE (deviation)	Stage 6/7 system model in Python, GNU Octave (PASS vs Python) and Scilab Xcos block diagram (0.04 % vs Octave); Simulink V20 run in MATLAB Online by the user: all 4 cases PASS vs Xcos (solar heat within 0.05 %)
12 Results and figures	MOSTLY DONE	See Section 3
13 Patent strategy	NOT ACTIONED	Engineering cross-check only; V1, V10 open
14 Schedule and risk	REFERENCE	—
15 Rev H corrections	DONE	Converging geometry (C1) used throughout; C7, C8 open (manufacturing dimensions)

2. Verification register V1–V10

#	Item	Status	Evidence / note
V1	Prior-art null results	FIRST PASS DONE	Web first pass (20_Patent_prior_art): no document found combining roll-bond + converging channels (+ TIM/PCM/VIP); inventive step weak-moderate; professional search on Espacenet/Patentscope/InPASS pending, only if filing
V2	COMSOL licence	NOT APPLICABLE	COMSOL not used
V3	MATLAB licence	DONE	Section 11 in GNU Octave, Scilab Xcos and MATLAB/Simulink; the user ran GRAIL_simulink_complete.m in MATLAB Online and reported the result as matching (PASS vs Xcos); V20 printed output archived in 17_Simulink_MATLAB_model/V20_run_record.md (all 4 cases PASS)
V4	COMSOL feature names	NOT APPLICABLE	OpenFOAM v1912; every solver option verified in the installed binary
V5	Baseline validation < 8 %	DONE (partial pass, method level)	16 papers screened (folder 11). Del Col 2013 method check (folder 19): heat loss within test uncertainty, η_0 over-predicted 6–8 %. Beikircher 2015 component check (folder 21): TIM front loss within measured range; catalogue VIP conductivity optimistic (a_1 2.1 $\rightarrow$ 2.3–2.5 W/m ² K). η_0 within 8 % (+6.0 / +7.8 %); curve within 8 % up to $T_m \cdot 0.02$ m ² K/W. No experimental validation of the present collector is claimed. Closed by the user; record: 11_Validation_V5/V5_FINAL_closure.md
V6	Supplier datasheets	OPEN	Literature properties used
V7	Counter-current effect on R4, matched mean T	DONE	GRAIL-CHT at matched plate mean 320/330/340 K (29_v7): $\Delta\eta = -0.001$ points, $\Delta R_4 = +0.04$ %, plate std +8 to +11 % for alternating; alternating needs ~15 K cooler inlet for the same plate temperature
V8	η_0 , a_1 , a_2 fit	DONE	Stage 6 ISO 9806 fit on 22 GRAIL-CHT runs: alternating η_0 0.597, a_1 2.00 W/m ² K; parallel η_0 0.637, a_1 2.10 W/m ² K; held-out errors ≤ 0.24 points
V9	TO geometry re-simulated	NOT APPLICABLE	No TO
V10	Patent-agent review	OPEN	—

3. Figures (roadmap 12.1)

#	Figure	Status	Delivered as
F1	Grid independence	DONE	F01 (3-D axial), F02 (cross-section Nu), F04 (ROM axial)
F2	Validation curve vs Gunjo/Sun	NOT POSSIBLE	Reference rejected; F10 Nu envelope and F05 ROM-vs-3-D instead
F3	Efficiency curves conventional vs GRAIL	PARTIAL	GRAIL curves done (F18, Stage 6); no conventional collector modelled
F4	Geometry family incl. TO	PARTIAL	CAD report figures; no TO
F5 / F8	Temperature contours; counter-current side by side	DONE	F09 3-D rear-surface maps, same scale
F6	Velocity field	DONE	CAD/CFD report figures S3, S6; videos
F7	Plate/fluid T along flow	DONE	F08
F9	Temperature histogram	DONE	F09b
F10	R4 vs bridge width	DONE	F10b (T_R4 - T_mean vs bridge)
F11	Loss breakdown	DONE	F11
F12	Correlation heatmap	DONE	F12
F13 / F14	Grading result and penalty	DONE	F13_F14

4. The three headline numbers (roadmap 12.3)

#	Roadmap sentence	Answer from the closed CFD phase
1	Counter-current reduced R4 by ___ % at unchanged mean T, raising η by ___ points	ANSWERED (V7): at matched mean plate temperature counter-current changes R4 by +0.04 % (not a reduction) and η by -0.001 points (no change); plate std is 8–11 % higher. At unchanged inputs alternating runs +13.9 K hotter and -7.3 % less efficient because it needs a ~15 K cooler inlet to reach the same plate temperature
2	TO absorber delivered ___ % more	NOT APPLICABLE — no TO
3	Annual kWh/m ² /year	DONE (Stage 7, Berhampur TMY): collector yield at 40 °C inlet 1,030 (parallel) / 965 (alternating) kWh/m ² /yr; POA 1,976 kWh/m ² /yr
Consequence for Filing 1		
The roadmap (15.8) warns that Filing 1 is a claim about a flatter temperature field. The corrected CFD shows a flatter field only at high flow; at matched mean plate temperature (V7) the arrangements have the same efficiency and alternating is slightly less uniform at the design flow. Claim language must match this evidence.		

5. Roadmap 15.8 corrected sequence

Step	Status
1 Re-export STEP with suppressions	DONE (as direct geometry evaluation)
2 Isothermal mass check within 0.1 %	DONE — mass imbalance $\sim 1\text{e-}9$
3 Reynolds 23.9–198.8	DONE — Rev 4.1 Re_in 21.8–151.3 (correct non-circular definition; low end below range because $\dot{m}$ bound 1.0 g/s)
4 Validation gate 8.7	PASSED at method level (V5); no test of the present collector
5 Variants; counter-current at matched mean T	DONE — see V7 (29_v7)
6 w_bridge 20–60 mm with r_fillet 0.5–2.0 mm	PARTIAL — 20–37 mm (geometric hard limit at 40 mm pitch); fillet not swept. DESIGN_REVIEW_REQUIRED

6. Remaining open items

- V5: closed at method/component level (Del Col 2013, Beikircher 2015). A prototype test would be needed for experimental validation of GRAIL itself. DESIGN_REVIEW_REQUIRED: use a measured in-collector VIP conductivity (0.010–0.018 W/m·K).
- V1 professional search and V10 patent-agent review, only if a filing is planned (first-pass search in 20_Patent_prior_art).
- V6: supplier datasheets (literature properties used).
- C7, C8: frame wall thickness and manifold dimensions — design inputs from the user.
- Bridge range above 37 mm (needs a different pitch) and the fillet sweep.
- Design decisions recorded: 2 modules is the system design (4 modules stagnate above 420 K — overheating protection needed for larger arrays); the NSGA-II optimum at the flow bound with $G \rightarrow 1$ means the converging channel is argued for plate uniformity, not efficiency.
- Deferred by choice: PCM latent heat, radiation / optical 3-D model, full 3-D campaign.